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Geographical distance, investment strategies and performance of venture capital: evidence from Chinese enterprises

Jiaoe Wang^{a,b}, Fengyuan Liu^c, Jianjun Li^d, Delin Du ^a and Jiafan Wu^d

ABSTRACT

A growing body of evidence suggests that geographical distance plays an important role for venture capital (VC) investment. This study explores whether and how the geographical distance between the lead VC firms (VCFs) and the invested entrepreneurial firms (IEFs) influences the investment strategies of VCFs and their performance. The results show: (1) The geographical distance has a significant positive effect on the performance of VC investments. (2) As distance increases, lead VCFs will tend to be conservative on investments by improving project screening criteria and adopting syndication strategies and multiple-rounds of staged financing, which can dramatically improve the successful exit from the invested projects. This study has useful theory and policy implications for VCFs, entrepreneurial firms and local governments.

KEYWORDS

Geographical distance, venture capital investment, performance, venture capital firms, invested entrepreneurial firm

JEL G24, R12

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1. INTRODUCTION

China is increasingly becoming an important venture capital (VC) market, with a total amount of 93.8 billion US dollars of venture investment in 2018, ranking first globally. The geography of China's venture capital firms (VCFs) is highly spatially concentrated in mega-cities such as Beijing, Shanghai and Hangzhou, which is consistent with that of developed countries (Martin et al., 2005). From 2015 to 2017, the amount of VC in those three cities accounted for more than 30% of the world VC market, surpassing the 27.6% market share of the three largest VC cities in the United States (San Francisco, New York and Boston). The agglomeration of VCFs geographically leads to excessive competition in high-quality projects in local cities, so diversifying the geography of investments has become popular to increase potential high-quality projects. Bowden et al. (2016) confirmed that investing in diversified regions can improve the return rate of VC funds. Therefore, VCFs have gradually conducted their investments across cities.

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Distance has greatly influenced VC investment. In general, the VC investment process consists of several steps, including deal origination, screening, due diligence and structuring in the pre-contractual phases, investment development and exit in the post-contractual phases (Bender, 2011). In the process, some information must be obtained through face-to-face meetings (Lutz et al., 2013). Within a short distance, VCFs can easily visit local start-ups and have frequent in-person contact with entrepreneurs. In practice, some VCFs even make their investment decisions based on the ‘twenty-minute rule’, which means if a startup enterprise seeking VC is not within a twenty-minute drive of a VC’s offices, it will not be funded. VC investment has local bias and it will be stronger when it acts as the lead VC and when it is investing alone, while the reputation of VCFs and VCs with broader networks exhibit less local bias (Cumming & Dai, 2010). In addition, the distribution of VCFs and invested entrepreneurial firms (IEFs) both show spatial agglomeration, and there is a certain overlap between them, such as the biotechnology industry in America (Powell et al., 2002).

Distance matters for the eventual performance of VC through shifting investment strategies and adding value service. Despite the spatial proximity preference and local bias, cross-regional VC investments gradually account for a large proportion of financial globalisation (Florida & Smith, 1993; Pan et al., 2016; Zhang, 2011). With the increase in distance, information asymmetry, cultural isolation and institutional environment differences will increase accordingly. Values and the judgment of IEFs are uncertain to some extent. Thus, VCFs usually will adopt a more cautious investment strategy such as staged financing and joint investment, to mitigate investment risk (Gu & Lu, 2014; Guler, 2007; Li, 2008). That is to say, geographical distance may not only significantly raise the difficulty of VCFs in selecting and supervising projects and value-added services, but also influence the stage preference, quality standards, syndicated investment, staged financing strategies of VCFs and ultimately affect the performance of VC investment. Given these strategies, we posit that there may be a positive correlation between the distance and the success rate of VCF exits.

However, most studies have analysed the influence of geographical distance on the probability and distribution of VC investment, and there is little literature concerning the influence of geographical distance on investment strategies of VCFs and VC performance. This paper will provide empirical evidence on whether and how geographical distance matters, with a large-scale empirical study based on the data of 21,580 rounds of VC deals in China from 2000 to 2016.

2. LITERATURE REVIEW

VC investments are high-risk financial activities, necessitating thorough project screening and evaluation to mitigate risks and enhance returns (Tyebjee & Bruno, 1984). This entails extensive information exchange between the VCF and IEF, with face-to-face communication playing a crucial role in acquiring implicit information (Fritsch & Schilder, 2008; Tian, 2011). The geographical distance between these entities is intricately linked to travel time and costs, thereby influencing information asymmetry and transaction expenses. These factors, in turn, impact VC investment strategies (Chemmanur et al., 2016), ultimately shaping investment performance. Therefore, in this section, we conducted a review of existing literature examining the correlation between geographical distance and VC investment, as well as the connection between VC investment strategies and the investment performance.

2.1. Geographical distance and VC investment

Impacts of distance on VC investments concern a lot of scholars, and most studies reveal the spatial proximity preference and local bias in the selection decision of VC investments. Cumming and Dai (2010) developed measures of local bias in VC investments and analysed the factors that contribute to local bias and linked the geographical distance to VC investments. Further, Lutz

et al. (2013) confirmed that the increase in distance between VCFs and IEFs significantly reduced the probability of VC investment. When assessing and choosing new investment prospects, diverse implicit knowledge is frequently necessary, typically gained through personal interactions (Fritsch & Schilder, 2008). Geographical distance can also impact the VCF's ability to oversee the start-ups and provide value-added services to them post-investment. Other than geographical distance, Tykvová and Schertler (2014) observed that institutional distance can also reduce the occurrence probability of transnational VC investment, while cooperation with local VCFs could decrease the negative impact of geographical distance but had no effect on institutional distance. For high-tech industry, geographical distance between VCFs and VC-backed enterprises significantly restrained the possibility of acquiring VC investment, and the possibility became zero if the distance was greater than 250 km (Colombo et al., 2019). As distance is increased, Kolympiris et al. (2018) revealed that patent activity and the founding team's entrepreneurial experience are more effective in increasing VC investments.

2.2. VC investment strategy and performance

The impact of distance on VC performance has attracted some studies. Belderbos et al. (2018) noted that the geographic diversification in corporate VC investments has positive consequences for firms' technological performance in the context of simultaneously pursued technology alliance strategies. Tian et al. (2020) explored whether and how the geographical distance between VCFs and IEFs affects the technological performance of VC-backed enterprises from the perspective of the VC's equity backgrounds and investment strategies.

Different strategies in VC investment processes have important impacts on the investment performance. Many studies have focused on the syndication and staged financing strategies and, in general, the two strategies are both effective at mitigating the principal-agent risk and project development risk (Gu & Lu, 2014; Li, 2008). Guler (2007) suggested that staged financing reduced the investment risk by multi-round financing and encouraged the IEF to achieve the milestone goal by improving itself. Syndication strategy is the joint investment in an entrepreneurial firm by multiple VCFs, which can reduce the selection risk of project screening, approach better projects and alleviate overall investment risk (Hochberg et al., 2007). Brander et al. (2002) confirmed that the syndicated investments have higher returns, favouring the value-added interpretation. There are also some other important approaches in the investment process to alleviate the risk of asymmetric information between the investors and investees, such as stock options, trading terms, convertible securities and board seats (Arcot, 2014). Chemmanur et al. (2016) confirmed that, compared with enterprises invested by purely international or purely local venture capitalists, enterprises backed by a syndication composed of international and local venture capitalists had more successful exits and higher post initial public offering (IPO) operation performance.

Owing to the collection of soft information, the screening of VC projects, post-investment supervision and value-added services in VC investments all require face-to-face communication between VCFs and entrepreneurial firms (Tian, 2011). The influence of geographical distance on travel time would inevitably affect the investment strategy and performance. However, existing studies mainly focus on the influence of geographical distance on the probability of VC investment, and there is little literature concerning the influence of geographical distance on investment strategies of VCFs and VC performance. Therefore, our study focuses on this issue and aims to reveal the relationship between geographical distance, investment strategies and VC investment performance.

3. RESEARCH FRAMEWORK AND HYPOTHESIS

As a high-risk financial activity, there are risks of information asymmetry in VC investments. In order to reduce the risks and transaction costs, and increase returns, VCFs usually adopt many

strategies in their investment procedures, including project screening and trading strategies, which influence the investment performance together with post-investment supervision and value-added services (Gompers et al., 2020; Tian, 2011). No matter what kind of investment criteria or investment strategy, both sides of the investment need to carry out frequent information and personnel exchanges during the investment process, which means that geographical distance would play a certain role. As shown in Figure 1, the theories of information asymmetry and transaction cost provide important support for our understanding of the relationship between geographical distance, investment strategies and VC investment performance.

The increase in distance generally affects the cost for VCF investment managers, which in turn increases the costs of information collection and project supervision, especially for high-risk projects. Contrasted with established or publicly listed enterprises, obtaining detailed information about startups is more challenging. This implies a potentially more serious information asymmetry between VCFs and new ventures (Gompers & Lerner, 2004). Closeness in geographical proximity often translates to convenient access to information and communication (Cumming & Dai, 2010; Lutz et al., 2013). Perhaps VCFs lower the bar on new investments if the marginal cost of visiting the company is reduced (Chen et al., 2010). In other words, if the distance is greater, VCFs may raise the screening criteria for new investments. On the one hand, they may reduce investment risk preference, which is reflected in the preference for later-stage projects. On the other hand, VCFs improve the quality requirements for more distant entrepreneurial firms to improve the quality of investment projects. Thus, we hypothesise the following:

H1: The greater the geographical distance between the lead VCF and the IEF, the more likely it is that the VCF will invest in later-stage projects to enhance the probability of a successful investment.

H2: The farther the geographical distance between the lead VCF and the IEF, the higher the quality requirements for IEFs to enhance the probability of a successful investment.

VCFs commonly engage in syndication when investing in new ventures. Previous research has demonstrated that syndication plays an important role in mitigating information asymmetry issues (Manigart et al., 2006; Wright & Lockett, 2003). Syndication can help VCFs improve the efficiency of discriminating IEFs, so diversified resources can be brought to alleviate the shortage of human and financial resources, and ultimately improve the efficiency of value-added services (Meuleman et al., 2009). Chen et al. (2010) have confirmed that syndication

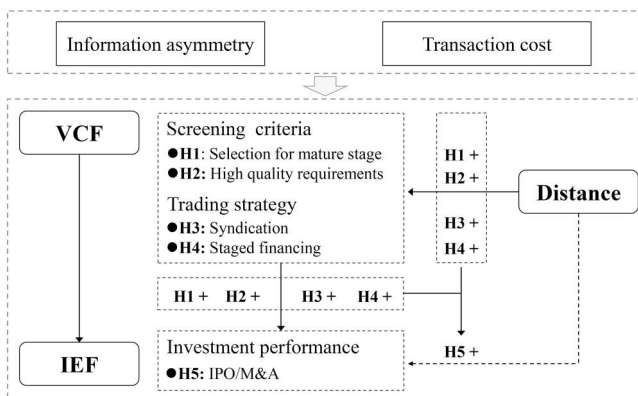


Figure 1. Research framework of geographical distance, investment strategy of VCF and VC performance.

networks can reduce the local bias of VCFs. Therefore, we expect that syndication is most likely related to the geographical distance in the following way:

H3: The farther the geographical distance between the leading VCF and the IEF, the more likely a VCF is to adopt the prudent trading strategy of syndication to improve the probability of a successful investment.

VCFs strategically stage their investments to address information asymmetries and agency costs. The adoption of a staged financing strategy allows VCFs to effectively supervise and motivate entrepreneurial teams (Tian, 2011). However, this staging approach comes with associated costs, such as contracting and negotiation expenses for each financing round. When VCFs anticipate a heightened need for monitoring, they are inclined to opt for more frequent staging. Consequently, in scenarios involving significant geographical distances, VCFs may employ a staged financing strategy to mitigate the challenges posed by limited cost budgets for investment project screening and post-investment supervision. Thus, we hypothesise the following:

H4: The greater the geographical distance between the VCF and the IEF, the more likely it is that the VCF will adopt a prudent trading strategy of staged investment to improve the probability of a successful investment.

In the above four hypotheses, we explore the screening criteria and trading strategy concerning geographical distance. The final hypothesis, in conjunction with the aforementioned considerations, associates distance with performance. Specifically, as the VCF moves farther away from the IEF, it is likely to heighten its project screening criteria and embrace a more conservative strategy to mitigate risk and improve the probability of a successful investment. In some existing literature, a distinction is made between the pure selection effect and treatment effect in empirical analyses (e.g., Bertoni et al., 2011; Brander et al., 2002; Sørensen, 2007). In this study, the focus is specifically on examining and referencing the pure selection effect. Thus, we put forward the following hypothesis:

H5: The farther the geographical distance between the leading VCF and the IEF, the higher the successful exit rate is for VCFs.

4. RESEARCH DESIGN AND DATA SOURCE

4.1. Variables and data

Table 1 shows the dependent, intermediate, explanatory and control variables used in this paper and their calculation methods. The data on VC investment rounds from 2000 to 2016 came from CVSource (<http://www.cvsource.com.cn/>). A total of 21,580 investment rounds were obtained, including 10,580 invested enterprises and more than 3000 VCFs. Since samples of local investment and non-local investment were included in this study, the geographical distance between the lead VCFs and IEFs belonging to the same prefecture-level city was defined as 0, and the distance between different cities was measured using ArcGIS. Statistics of variables are shown in Appendix A of the online supplemental data.

4.2. Model specifications

We employed the following logistic model to reveal the influence of distance on the performance of VC investment:

$$p(IPOMA = i|X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \ln(\text{distance}) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (1)$$

Table 1. Variables and descriptions.

Dependent variables	Description
IPOMA	If the IEF completes an IPO or is acquired during a five-year observation period after the investment, the value equals 1; otherwise it equals 0.
IPO	If the IEF completes an IPO during a five-year observation period after the investment, the value equals 1; otherwise it equals 0.
Independent variables	
Ln(distance)	The natural logarithm of the distance (the shortest road time matched according to the road network of the year of investment) between the city where the IEF is located and the city where the lead VCF is located.
Ln(flying distance)	The natural logarithm of the geometric distance between the city where the entrepreneurial firm is located and the city where the lead VCF is located.
Ln(road distance)	The natural logarithm of the road distance (the shortest road distance matched according to the road network of the year of investment) between the city where the IEF is located and the city where the lead VCF is located.
Intermediate variables	
stage	If the IEF is in an early stage or development stage, the value is 1; otherwise it is 0.
A-round	If the investment belongs to the angel or A-round, the value is 1; otherwise it is 0.
Ln(patent-3)	The natural logarithm of the total number of invention patents in the three years before the investment.
Ln(patent-5)	The natural logarithm of the total number of invention patents in the five years before the investment.
Ln(patent + 3)	The natural logarithm of the total number of invention patents in the three years after the investment.
Ln(patent + 5)	The natural logarithm of the total number of invention patents in the five years after the investment.
Syndication size	The total number of VCFs involved in syndication in the entrepreneurial firm.
Ln(round duration)	The natural logarithm of the average monthly interval of each round of IEF financing after this round of investment
Round sum	The total number of rounds of IEF financing after this round of investment
Ln(average round investment)	The natural logarithm of the average investment amount of each round of IEF subsequent rounds after this round of investment
Control variables	
IEF location	If the IEF is located in Beijing, Shanghai and Shenzhen, the value is 1; otherwise it is 0.
Ln(lead VCF investment)	The natural logarithm of the investment amount of the leading VCF in this round of investment (RMB 10,000).
Ln(lead VCF experience)	The natural logarithm of the total number of months from the establishment of the lead VCF to the time when the investment occurs.
EPU	China's economic policy uncertainty index in the year of the investment (Baker et al., 2016)

(Continued)

Table 1. Continued.

Dependent variables	Description
GPR	The geopolitical risk index in the year of the investment (Caldara & Iacoviello, 2022)
HSR	If the city where the VCF is located and the city where the IEF is located have opened high-speed train in the investment year, the value is 1; Otherwise take 0
GDPper	The natural logarithm of GDP per capita of the province where the IEF is located
IPOA5	The average number of A-share IPOs per year in the five years following the investment year
Market	The marketisation index of the province where IEF is located in the year of investment (Wang et al., 2019).

where $p(IPOMA = i|X, Z)$ is the probability of IPO or mergers and acquisitions (M&A) of the entrepreneurial firm during a five-year observation period after the investment, since IPO or M&A dummy variables are generally used to measure the performance of VCFs (Chemmanur et al., 2016; Dimov & Shepherd, 2005; Hochberg et al., 2007; Nahata, 2008). Explanatory variable X is the distance between the lead VCF and the IEF. Z_{it} is the vector of the control variables, which include IEF attributes such as location, and VCF attributes such as experience and investment amount, and the six variables EPU, GPR, HSR, GDPper, IPOA5, Market.

To investigate the mechanisms by which geographical distance influences investment performance, we utilised the causal steps approach developed by Baron and Kenny (1986) to validate the screening criteria and trading strategies posited earlier.

We employ two variables – stage (whether the IEF is in the early stage) and A-round (whether the investment round belongs to the A round and the angel round) – to gauge the stage risk appetite when the VCF invests in projects with a longer geographical distance, thereby substantiating H1. The subsequent logit regression is employed for this purpose:

$$stage = \frac{1}{1 + e^{-[\alpha + \beta \ln(distance) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (2)$$

$$A - round = \frac{1}{1 + e^{-[\alpha + \beta \ln(distance) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (3)$$

$$p(IPOMA = i|X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \ln(distance) + \theta stage + Z'_{it} \gamma + \epsilon_{it}]}} \quad (4)$$

$$p(IPOMA = i|X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \ln(distance) + \theta A - round + Z'_{it} \gamma + \epsilon_{it}]}} \quad (5)$$

Since the number of patents can effectively reflect the innovation capacity of the enterprises and the likelihood of future success, it serves as a suitable metric. Therefore, the count of the invention patents in the 3 years and 5 years preceding the investment is employed to assess the quality of the IEF. Additionally, an IEF with robust innovation capabilities is anticipated to experience a sustained growth in the number of patents following the investment. Consequently, we also gauge the quality of the IEF by considering the count of invention patents 3

years and 5 years post-investment. The following models are utilised:

$$\text{Ln}(\text{patent} - 3) = \alpha + \beta \text{Ln}(\text{distance}) + Z'_{it} \gamma + \epsilon_{it} \quad (6)$$

$$\text{Ln}(\text{patent} - 5) = \alpha + \beta \text{Ln}(\text{distance}) + Z'_{it} \gamma + \epsilon_{it} \quad (7)$$

$$\text{Ln}(\text{patent} + 3) = \alpha + \beta \text{Ln}(\text{distance}) + Z'_{it} \gamma + \epsilon_{it} \quad (8)$$

$$\text{Ln}(\text{patent} + 5) = \alpha + \beta \text{Ln}(\text{distance}) + Z'_{it} \gamma + \epsilon_{it} \quad (9)$$

$$p(\text{IPOMA} = i | X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \text{Ln}(\text{distance}) + \theta \text{Ln}(\text{patent} - 3) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (10)$$

$$p(\text{IPOMA} = i | X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \text{Ln}(\text{distance}) + \theta \text{Ln}(\text{patent} - 5) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (11)$$

$$p(\text{IPOMA} = i | X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \text{Ln}(\text{distance}) + \theta \text{Ln}(\text{patent} + 3) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (12)$$

$$p(\text{IPOMA} = i | X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \text{Ln}(\text{distance}) + \theta \text{Ln}(\text{patent} + 5) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (13)$$

Finally, we scrutinise the mechanism of the investment strategy. This analysis involves four variables: syndication size (the overall number of VCFs participating in the investment), round duration (the average monthly interval of each round of IEF financing after this specific investment round), investment rounds (the total number of IEF financing rounds following this specific investment round) and average round investment (the average investment amount of each IEF round subsequent to this specific investment round). The aim is to validate both H3 and H4. The models are as follows:

$$\text{Syndication size} = \alpha + \beta \text{Ln}(\text{distance}) + Z'_{it} \gamma + \epsilon_{it} \quad (14)$$

$$\text{Ln}(\text{round duration}) = \alpha + \beta \text{Ln}(\text{distance}) + Z'_{it} \gamma + \epsilon_{it} \quad (15)$$

$$\text{Round sum} = \alpha + \beta \text{Ln}(\text{distance}) + Z'_{it} \gamma + \epsilon_{it} \quad (16)$$

$$\text{Ln}(\text{average round investment}) = \alpha + \beta \text{Ln}(\text{distance}) + Z'_{it} \gamma + \epsilon_{it} \quad (17)$$

$$p(\text{IPOMA} = i | X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \text{Ln}(\text{distance}) + \theta \text{Syndication size} + Z'_{it} \gamma + \epsilon_{it}]}} \quad (18)$$

$$p(\text{IPOMA} = i | X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \text{Ln}(\text{distance}) + \theta \text{Ln}(\text{round duration}) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (19)$$

$$p(\text{IPOMA} = i | X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \text{Ln}(\text{distance}) + \theta \text{Round sum} + Z'_{it} \gamma + \epsilon_{it}]}} \quad (20)$$

$$p(\text{IPOMA} = i | X, Z) = \frac{1}{1 + e^{-[\alpha + \beta \text{Ln}(\text{distance}) + \theta \text{Ln}(\text{average round investment}) + Z'_{it} \gamma + \epsilon_{it}]}} \quad (21)$$

4.3. Spatial distribution of VC investment in China

Based on the panel data of VC investment from 2000 to 2016, we found that Chinese VCFs are mainly aggregated in the top five cities: Beijing, Shanghai, Shenzhen, Hangzhou and Tianjin (Figure 2). Almost half of inter-city investment in the last decade was from these cities. Moreover, the proportion of non-local investment rounds to all investment rounds increased

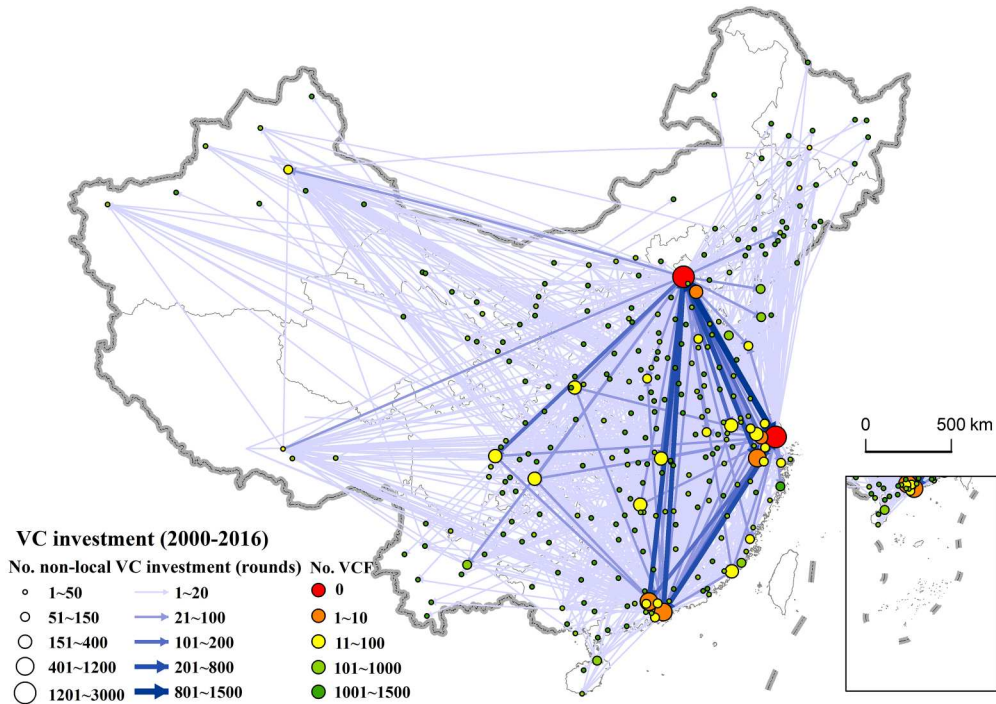


Figure 2. Spatial distribution of non-local VC investment flow and the agglomeration of VCFs.

significantly, from 40.4% in 2000 to 63.3% in 2016 (Figure 3). The above analysis shows that spatial over-gathering of VCFs has led to a gradual increase in the proportion of non-local investment, with it reaching a considerable level. Considering that investment managers have time budget constraints, geographical distance has become an important influencing factor for VCFs to conduct non-local investment. The impact of this distance on investment strategy and efficiency is investigated next.

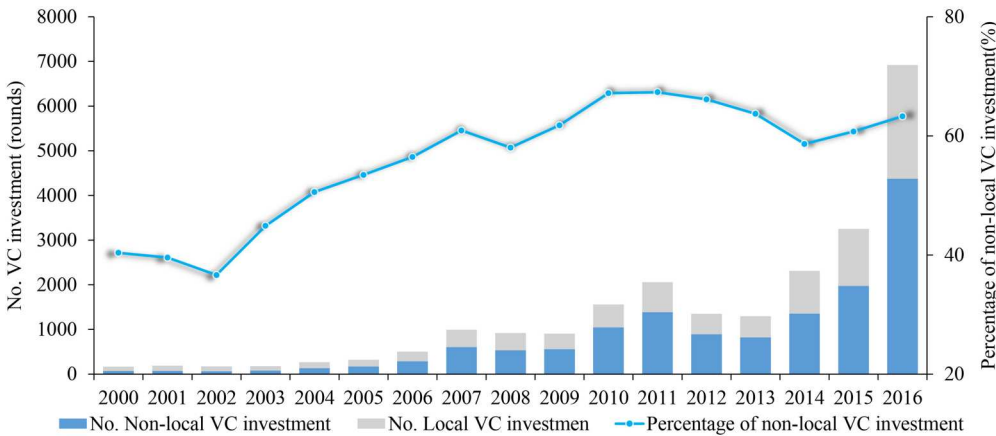


Figure 3. Increase in non-local VC financing by investment round in China.

5. EMPIRICAL RESULTS

5.1. Geographical distance and VC investment performance

We examined the impact of distance on the successful exit of the VCF from the investment (Table 2). Models (1) and (2) adopted multiple linear logit regression and used the distance between the lead VCF and the IEF for regression analysis on the probability of a successful IPO or M&A of the IEF. The empirical results of Model (1) fixed the effects of industry, time and province. On this basis, Model (2) further adds control variables, including the characteristics of IEF, VCF and cities. The results of both models show that distance has a positive effect on IPOMA. This implies that during the 5-year observation period, a greater distance between the lead VCF and IEF correlates significantly with an increased likelihood of a successful IPO or M&A for the IEF. When the distance increases by 1 km, the successful exit rate of the VCF increases by 5.38%.

Table 2. Regression of the distance on the success rate of VC investments.

	(1) IPOMA	(2) IPOMA
Ln(distance)	0.0973*** (5.72)	0.0538*** (3.19)
IEF location		−0.4462*** (−3.58)
Ln(lead VCF investment)		0.2809*** (20.51)
Ln(lead VCF experience)		−0.0664*** (−4.43)
EPU		0.0007 (1.16)
Market		−0.1229** (−2.39)
HSR		−0.0700 (−1.08)
GDPper		−0.0000 (−1.59)
IPOA5		−0.0099*** (−8.50)
Industry fixed effects	yes	yes
Year fixed effects	yes	yes
Province fixed effects	yes	yes
Constant	−0.3173 (−1.61)	0.3169 (0.77)
N	25447	25447
r ² _p	0.1314	0.1476
chi ²	3.3e + 03	3.2e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3. Robustness test of the distance on the success rate of VC investments.

	(3) IPOMA	(4) IPOMA	(5) IPO
Ln(flying distance)	0.0277*** (3.83)		
Ln(road distance)		0.0254** (3.59)	
Ln(distance)			0.0467** (2.25)
IEF location	−0.4229*** (−3.39)	−0.5216*** (−3.79)	−0.5359*** (−3.89)
Ln(lead VCF investment)	0.2800*** (20.45)	0.2807*** (16.81)	0.2813*** (16.85)
Ln(lead VCF experience)	−0.0666*** (−4.45)	−0.0432** (−2.41)	−0.0431** (−2.41)
EPU	0.0007 (1.14)	0.0008 (1.14)	0.0008 (1.16)
Market	−0.1219** (−2.37)	−0.1075* (−1.75)	−0.1072* (−1.75)
HSR	−0.0629 (−0.97)	−0.0871 (−1.16)	−0.0913 (−1.21)
GDPper	−0.0000* (−1.76)	−0.0000 (−0.10)	−0.0000 (−0.00)
IPOA5	−0.0099*** (−8.56)	−0.0097*** (−7.12)	−0.0096*** (−7.10)
Industry fixed effects	yes	yes	yes
Year fixed effects	yes	yes	yes
Province fixed effects	yes	yes	yes
Constant	0.2866 (0.70)	−0.7842 (−1.63)	−0.7672 (−1.59)
N	25447	25447	25447
r2_p	0.1477	0.1810	0.1809
chi2	3.2e + 03	2.9e + 03	2.9e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The robustness of this discovery is explored by substituting the distance variable with alternative measures to evaluate its influence on VC performance. Specifically, we employ the flying distance and road distance as substitute variables, controlling for relevant factors, as depicted in Table 3. Models (3) and (4) reveal that both the flying distance and road distance exhibit a significant positive impact on IPOMA. Furthermore, we scrutinise the impact of distance when considering only IPO as an indicator of investment performance. As demonstrated in Model (5), distance maintains a notable positive effect when defining successful exits solely in terms of IPOs. All these analyses substantiate the robust positive effect of distance on VC performance.

5.2. Geographical distance and screening criteria

To analyse why geographical distance is significantly positively correlated with successful investments, this section focuses on the screening criteria of VCFs, i.e., the stage preference and quality requirements.

5.2.1. Correlations between distance and stage preference

To analyse the stage preference for distant IEFs, we defined the investment projects in the early stage or development stage as a dummy variable. After controlling other related variables, the distance variable is significantly negatively correlated with the early-stage dummy variable of IEFs as shown in Model (6) in Table 4. This suggests that the farther the geographical distance, the less willing VCFs are to invest in early-stage projects. Further examination was carried out on series A and preceding investments, and the regression outcomes are depicted in Model (7). The finding reveals that distance significantly exerts a negative impact, thereby confirming that an increased distance correlates with a diminished inclination for A-round and angel round investments.

Incorporating the stage preference into the model examining the impact of distance on VC performance, as presented in Table 5. Both early-stage investments and series A and angel

Table 4. Regression of distance on investment stage preference.

	(6) Stage	(7) A-round
Ln(distance)	−0.1101*** (−6.19)	−0.0723*** (−5.49)
IEF location	0.1426 (0.89)	0.2410** (2.29)
Ln(lead VCF investment)	−0.3697*** (−21.63)	−0.2379*** (−23.44)
Ln(lead VCF experience)	−0.0080 (−0.50)	0.0205* (1.73)
EPU	0.0015* (1.71)	−0.0001 (−0.31)
Market	−0.1679*** (−2.85)	0.0501 (1.18)
HSR	0.0664 (0.94)	−0.0205 (−0.39)
GDPper	−0.0000*** (−3.95)	−0.0000 (−0.72)
Industry fixed effects	yes	yes
Year fixed effects	yes	yes
Province fixed effects	yes	yes
Constant	5.8242*** (10.64)	3.1076*** (7.68)
N	25447	25447
r ² _p	0.1950	0.1405
chi ²	3.2e + 03	4.9e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5. Regression of distance, VC stage preference and investment performance.

	(8) IPOMA	(9) IPOMA
Ln(distance)	0.0314* (1.83)	0.0377** (2.20)
Stage	−0.9149*** (−17.91)	
A-round		−0.9151*** (−21.09)
IEF location	−0.4597*** (−3.65)	−0.4356*** (−3.47)
Ln(lead VCF investment)	0.2404*** (17.21)	0.2172*** (15.70)
Ln(lead VCF experience)	−0.0688*** (−4.53)	−0.0681*** (−4.50)
EPU	0.0008 (1.34)	0.0007 (1.27)
Market	−0.1401*** (−2.68)	−0.1197** (−2.29)
HSR	−0.0449 (−0.68)	−0.0805 (−1.23)
GDPper	−0.0000** (−2.44)	−0.0000** (−2.21)
IPOA5	−0.0122*** (−10.32)	−0.0113*** (−9.66)
Industry fixed effects	yes	yes
Year fixed effects	yes	yes
Province fixed effects	yes	yes
Constant	1.8192*** (4.28)	1.6956*** (4.02)
N	25447	25447
r2_p	0.1625	0.1692
chi2	3.5e + 03	3.6e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

rounds exhibit a significant negative effect on the successful exit rate of VCF. This suggests that relatively mature projects tend to have a higher success rate. In both Models (8) and (9), distance retains a positive influence on the successful exit rate of the VCF. This indicates that, with greater distance, VCFs are inclined to choose relatively mature projects, resulting in a higher success rate, thus confirming H1.

5.2.1. Correlations between distance and quality requirements

We utilise invention patents as a metric for characterising the quality of IEF. The outcomes presented in Table 6 indicate a positive correlation between distance and the number of invention

Table 6. Regression of distance on quality of IEF.

	(10) Ln(patent-3)	(11) Ln(patent-5)	(12) Ln(patent + 3)	(13) Ln(patent + 5)
Ln(distance)	0.0080 (1.22)	0.0144* (1.90)	0.0117 (1.63)	0.0166** (2.11)
IEF location	-0.1487*** (-2.74)	-0.1234* (-1.96)	-0.1382** (-2.31)	-0.1192* (-1.83)
Ln(lead VCF investment)	0.0665*** (13.44)	0.0741*** (12.90)	0.0726*** (13.32)	0.0760*** (12.80)
Ln(lead VCF experience)	0.0069 (1.13)	0.0059 (0.83)	0.0076 (1.14)	0.0026 (0.36)
EPU	-0.0002 (-1.09)	-0.0002 (-1.15)	-0.0002 (-1.23)	-0.0003 (-1.23)
Market	0.0065 (0.29)	0.0005 (0.02)	0.0077 (0.32)	0.0014 (0.05)
HSR	-0.0350 (-1.25)	-0.0516 (-1.59)	-0.0426 (-1.39)	-0.0543 (-1.62)
GDPper	0.0000 (0.93)	0.0000 (0.90)	0.0000 (0.97)	0.0000 (0.91)
Industry fixed effects	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes
Province fixed effects	yes	yes	yes	yes
Constant	-0.5750*** (-3.07)	-0.6185*** (-2.84)	-0.6466*** (-3.13)	-0.5942*** (-2.64)
N	25447	25447	25447	25447
r ² _a	0.1016	0.1008	0.1035	0.0981
F	42.7163	42.3624	43.5844	41.1063

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

patents of the IEF, both 5 years before and after the investment deal. When the time window is narrowed to 3 years, distance also exhibits a positive correlation with IEF patents, though it doesn't achieve statistical significance. Nevertheless, the regression findings suggest that, as distance increases, VCFs are more inclined to select higher-quality IEFs.

The results in Table 7 further substantiate that, even when factoring in the quality of IEF, distance maintains a significant positive effect on the successful exit rate of the VCF. Models (14) to (17) illustrate that IEFs with more invention patents are more likely to eventually go public or be acquired. This validates H2, asserting that with greater distance, VCFs tend to select the high-quality projects, thereby enhancing their successful exit rate.

5.3. Geographical distance and trading strategies

With distant investments, in addition to reducing the risk preference, lead VCFs may use syndication and staged strategies to attenuate the lack of time for project screening, supervision and value-added services after investment.

Table 7. Regression of distance, quality of IEF and investment performance.

	(14) IPOMA	(15) IPOMA	(16) IPOMA	(17) IPOMA
Ln(distance)	0.0511*** (3.00)	0.0495*** (2.89)	0.0502*** (2.94)	0.0490*** (2.86)
Ln(patent-3)	0.2912*** (20.33)			
Ln(patent-5)		0.2688*** (21.80)		
Ln(patent + 3)			0.2762*** (21.20)	
Ln(patent + 5)				0.2639*** (22.21)
IEF location	-0.4270*** (-3.37)	-0.4451*** (-3.50)	-0.4377*** (-3.44)	-0.4487*** (-3.52)
Ln(lead VCF investment)	0.2651*** (19.01)	0.2662*** (19.06)	0.2656*** (19.02)	0.2665*** (19.08)
Ln(lead VCF experience)	-0.0683*** (-4.50)	-0.0685*** (-4.49)	-0.0689*** (-4.53)	-0.0677*** (-4.44)
EPU	0.0008 (1.37)	0.0008 (1.43)	0.0008 (1.42)	0.0009 (1.45)
Market	-0.1323** (-2.55)	-0.1313** (-2.52)	-0.1336** (-2.57)	-0.1317** (-2.52)
HSR	-0.0608 (-0.92)	-0.0570 (-0.86)	-0.0593 (-0.90)	-0.0569 (-0.86)
GDPper	-0.0000 (-1.49)	-0.0000 (-1.50)	-0.0000 (-1.48)	-0.0000 (-1.50)
IPOA5	-0.0108*** (-9.23)	-0.0107*** (-9.13)	-0.0107*** (-9.15)	-0.0107*** (-9.07)
Industry fixed effects	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes
Province fixed effects	yes	yes	yes	yes
Constant	0.6263 (1.51)	0.6238 (1.50)	0.6361 (1.53)	0.6095 (1.46)
N	25447	25447	25447	25447
r2_p	0.1663	0.1693	0.1681	0.1702
chi2	3.6e + 03	3.6e + 03	3.6e + 03	3.6e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.3.1. Correlations between distance and syndication strategy

The diversified resources brought by cooperating with more VCFs can compensate for the shortage of human resources and market resources in VC investment (Deli & Santhanakrishnan, 2010) and ultimately improve the performance of the value-added services of the VCFs

(Hopp & Rieder, 2011). Under circumstances where the VCF is distant from the entrepreneurial firm, the time cost of information collection in project screening and face-to-face supervision increases significantly. It is extremely unfavourable for VCFs to improve screening quality and strengthen post-investment management but failing to do so may increase the moral hazard behaviour of entrepreneurial teams without supervision. At the same time, due to the increase in time cost caused by distance, value-added services may be limited. In the case of a significant increase in geographical distance, lead VCFs invite more VCs to participate in investment, which can improve screening performance and reduce information asymmetry with IEFs. More importantly, the risks of investment in different places can be dispersed. As indicated by Model (18) in Table 8, the distance variable exhibits a significantly positive correlation with the number of

Table 8. Regression of distance and trading strategies.

	(18) Syndication size	(19) Ln(round duration)	(20) Round sum	(21) Ln(average round investment)
Ln(distance)	0.0329*** (4.72)	0.0185** (2.04)	0.0227*** (3.89)	0.0769*** (3.60)
IEF location	0.0486 (0.84)	-0.1757** (-2.11)	0.1504*** (3.11)	0.5730*** (3.24)
Ln(lead VCF investment)	0.0741*** (14.11)	-0.0413*** (-5.76)	0.0283*** (6.42)	0.2063*** (12.79)
Ln(lead VCF experience)	0.0477*** (7.40)	0.0018 (0.21)	-0.0240*** (-4.43)	-0.0749*** (-3.79)
EPU	0.0001 (0.44)	0.0002 (0.56)	0.0001 (0.40)	0.0006 (1.03)
Market	0.0048 (0.21)	0.0089 (0.29)	0.0338* (1.72)	0.0811 (1.13)
HSR	-0.0327 (-1.10)	0.0473 (1.11)	0.0289 (1.16)	0.0765 (0.84)
GDPper	0.0000*** (6.32)	0.0000 (1.33)	0.0000*** (4.97)	0.0000*** (7.37)
Constant	0.5225*** (2.63)	7.8346*** (30.13)	0.3569** (2.14)	1.1575* (1.89)
Industry fixed effects	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes
Province fixed effects	yes	yes	yes	yes
N	25447	25447	25447	25447
r2_a	0.0774	0.2385	0.0604	0.0671
F	31.9210	39.7776	24.6885	27.5232

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

VCFs participating alongside each lead VCF. The findings from Model (22) in Table 9 further affirms that both syndication and distance exert a significant positive influence on the investment success rate. This supports the validity of H3.

Table 9. Regression of distance, trading strategies and investment performance.

	(22) IPOMA	(23) IPOMA	(24) IPOMA	(25) IPOMA
Ln(distance)	0.0460*** (2.71)	0.0232 (0.92)	0.0465*** (2.72)	0.0461*** (2.70)
Syndication size	0.2159*** (15.84)			
Ln(round duration)		−0.2130*** (−7.53)		
Round sum			0.3343*** (20.52)	
Ln(average round investment)				0.0976*** (20.61)
IEF location	−0.4618*** (−3.68)	−0.7240*** (−3.47)	−0.5192*** (−4.12)	−0.5226*** (−4.13)
Ln(lead VCF investment)	0.2702*** (19.57)	0.2503*** (11.62)	0.2769*** (19.97)	0.2637*** (18.95)
Ln(lead VCF experience)	−0.0751*** (−4.99)	−0.0374 (−1.63)	−0.0595*** (−3.93)	−0.0616*** (−4.07)
EPU	0.0007 (1.12)	0.0016 (1.56)	0.0006 (1.05)	0.0005 (0.91)
Market	−0.1251** (−2.42)	−0.0082 (−0.10)	−0.1354*** (−2.58)	−0.1319** (−2.53)
HSR	−0.0605 (−0.93)	−0.1692 (−1.56)	−0.0824 (−1.26)	−0.0807 (−1.23)
GDPper	−0.0000** (−2.02)	−0.0000** (−2.20)	−0.0000** (−2.18)	−0.0000** (−2.47)
IPOA5	−0.0096*** (−8.18)	−0.0150*** (−7.49)	−0.0086*** (−7.25)	−0.0083*** (−7.05)
Industry fixed effects	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes
Province fixed effects	yes	yes	yes	yes
Constant	0.1342 (0.32)	3.2025*** (4.47)	0.0609 (0.14)	0.0497 (0.12)
N	25447	25447	25447	25447
r2_p	0.1588	0.1855	0.1676	0.1673
chi2	3.4e + 03	1.7e + 03	3.6e + 03	3.6e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.3.2. Correlations between distance and staged financing strategy

Staged financing is an important VC investment strategy that gives VCFs an option to reduce risk by setting a milestone goal for the IEF, and only after the IEF achieves the goals set in advance will the VCF make subsequent investments, working as an effective incentive (Wang & Zhou, 2004) partially replacing the regulatory mechanism (Tian, 2011). After controlling for other influencing factors, the results of the seven multivariate linear regression Models (19) to (21) in Table 8 show that distance is significantly positively correlated with the three variables for the staged financing strategy. Moreover, we incorporated the staged financing strategy into the model assessing the impact of distance on VC performance. Models (23) to (25) in Table 9 substantiate that the success rate of investment experiences an increase to a certain extent with the reduction of time interval between different rounds, the rise in investment rounds and the average amount of investment. Therefore, lead VCFs tend to use a staged financing strategies to exert their own influence on entrepreneurial firms – significantly increasing the subsequent financing rounds of IEFs to control the investment risk and mitigate the moral risk of IEFs in long-distance investment. In summary, these results collectively corroborate H4.

6. DISCUSSION

The impact of distance on the successful exit rate of VCF is moderated by other factors. The reputation and experience of VCFs are often considered important factors in VC investment. As shown in Model (26) in Table 10, the experience of the leading VCF exhibits a significant negative moderating effect on the influence of distance on the successful exit rate of the VCF. Further analysis is conducted on the influence of the lead investment VCF experience on investment stage preference at a distance, and the results are presented in Table 11. The outcomes of Models (30) and (31) reveal that the experience of the lead VCF exerts a positive moderating effect on the impact of distance on early-stage investments. These findings indicate that more experienced VCs are more inclined to invest in early-stage IEFs, particularly in investments located at a greater distance. Given their heightened risk perception and resistance (Parhankangas & Hellström, 2007), experienced VCFs find it easier to embrace early-stage projects with elevated risks. This elucidates the significantly negative regression coefficient of $\text{Ln}(\text{distance})$ *lead VCF experience in Model (26). More experienced VCFs tend to select projects at an earlier stage in investments situated farther away, leading to investments in projects with greater risks and lower probabilities of a successful exit.

As demonstrated by Models (27) and (28) in Table 10, both EPU (reflecting economic policy uncertainty) and GPR (representing geopolitical risk) exhibit a significant negative influence on the relationship between distance and investment success. These findings suggest that heightened economic policy uncertainty and increased geopolitical risk contribute to a decreased likelihood of investment success for ventures located at a greater geographical distance. This correlation aligns with expectations, as greater economic policy uncertainty and geopolitical risks signify elevated investment risks.

Regarding the market aspect, Model (29) in Table 10 reveals that market development exerts a significant negative moderating impact on distance. This implies that market development diminishes the likelihood of successful exits for investments situated at a considerable distance. To explain this observation, we delve into the role of the market in the relationship between distance and investment stage preference, as depicted in Table 12. Models (32) and (33) demonstrate that the regression coefficient of the interaction term $\text{Ln}(\text{distance})$ *market is positive and achieves statistical significance. This occurs because higher levels of marketisation (market) lead VCFs to prefer early-stage IEFs. Due to the heightened risk associated with early projects, the exit rate of VCF results tends to decrease to some extent. This clarification elucidates why the

Table 10. Regression of moderating factors between distance and VC performance.

	(26) IPOMA	(27) IPOMA	(28) IPOMA	(29) IPOMA
Ln(distance)	0.1354*** (2.81)	0.1418*** (3.87)	0.1629*** (2.65)	0.5933*** (6.58)
Ln(distance)* Ln(lead VCF experience)	-0.0188* (-1.81)			
Ln(distance)*EPU		-0.0008*** (-2.71)		
Ln(distance)*GPR			-0.2477* (-1.85)	
Ln(distance)*market				-0.0642*** (-6.12)
IEF location	-0.4483*** (-3.59)	-0.4260*** (-3.41)	-0.4265*** (-3.41)	-0.4150*** (-3.32)
Ln(lead VCF investment)	0.2811*** (20.52)	0.2806*** (20.48)	0.2809*** (20.50)	0.2815*** (20.50)
Ln(lead VCF experience)	-0.0350 (-1.53)	-0.0660*** (-4.41)	-0.0662*** (-4.42)	-0.0631*** (-4.20)
EPU	0.0007 (1.19)	0.0024*** (2.81)	0.0006 (1.11)	0.0007 (1.16)
Market	-0.1239** (-2.41)	-0.1252** (-2.43)	-0.1229** (-2.39)	-0.0253 (-0.47)
HSR	-0.0658 (-1.01)	-0.0809 (-1.24)	-0.0757 (-1.16)	-0.0704 (-1.08)
GDPper	-0.0000 (-1.56)	-0.0000* (-1.73)	-0.0000* (-1.71)	-0.0000** (-2.56)
IPOA5	0.0072*** (2.94)	0.0069*** (2.84)	0.0071*** (2.91)	0.0071*** (2.90)
GPR	-6.9316*** (-6.51)	-6.7656*** (-6.34)	-6.3242*** (-5.67)	-6.8565*** (-6.44)
Industry fixed effects	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes
Province fixed effects	yes	yes	yes	yes
Constant	0.1135 (0.27)	0.0911 (0.22)	0.0249 (0.06)	-0.4731 (-1.10)
N	25447	25447	25447	25447
r2_p	0.1478	0.1480	0.1478	0.1494
chi2	3.2e + 03	3.2e + 03	3.2e + 03	3.2e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 11. The moderating effect of lead VCF experience on the relationship between distance and VCF stage preference.

	(30) Stage	(31) A-round
Ln(distance)	−0.2128*** (−3.99)	−0.1014*** (−2.60)
Ln(distance)* Ln(lead VCF experience)	0.0225** (2.04)	0.0066 (0.79)
IEF location	0.1397 (0.87)	0.2421** (2.30)
Ln(lead VCF investment)	−0.3701*** (−21.64)	−0.2381*** (−23.45)
Ln(lead VCF experience)	−0.0433* (−1.84)	0.0091 (0.48)
EPU	0.0015* (1.67)	−0.0001 (−0.32)
Market	−0.1700*** (−2.88)	0.0505 (1.19)
HSR	0.0656 (0.92)	−0.0227 (−0.43)
GDPper	−0.0000*** (−3.84)	−0.0000 (−0.73)
Industry fixed effects	yes	yes
Year fixed effects	yes	yes
Province fixed effects	yes	yes
Constant	5.9960*** (10.81)	3.1530*** (7.71)
N	25447	25447
r ² _p	0.1953	0.1405
chi ²	3.2e + 03	4.9e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 12. The moderating effect of market on the relationship between distance and VCF risk preference.

	(32) Stage	(33) A-round
Ln(distance)	−0.3054*** (−3.29)	−0.4063*** (−5.35)
Ln(distance)*market	0.0236** (2.15)	0.0390*** (4.47)
IEF location	0.1425 (0.89)	0.2150** (2.04)

(Continued)

Table 12. Continued.

	(32) Stage	(33) A-round
Ln(lead VCF investment)	−0.3703*** (−21.64)	−0.2373*** (−23.38)
Ln(lead VCF experience)	−0.0093 (−0.58)	0.0188 (1.58)
EPU	0.0015* (1.70)	−0.0001 (−0.31)
Market	−0.2088*** (−3.37)	−0.0151 (−0.34)
HSR	0.0617 (0.87)	−0.0180 (−0.34)
GDPper	−0.0000*** (−3.34)	−0.0000 (−0.02)
Industry fixed effects	yes	yes
Year fixed effects	yes	yes
Province fixed effects	yes	yes
Constant	6.1024*** (10.83)	3.5828*** (8.52)
N	25447	25447
r ² _p	0.1953	0.1410
chi ²	3.2e + 03	4.9e + 03

Note: t statistics in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

market index exhibits a negative moderating effect on the impact of distance on the successful exit rate of the VCF.

7. CONCLUSIONS

Existing studies have confirmed that VCFs tend to localise their investment preferences due to time costs and other reasons (Chen et al., 2010; Cumming & Dai, 2010; Lutz et al., 2013). However, the spatial agglomeration of VCFs leads to excessive competition in the screening of high-quality projects for local investment. Therefore, diversifying investment areas and carrying out non-local and even long-distance venture investment have become an effective investment strategy for most VCFs (Bowden et al., 2016). Inevitably, the significant increase in geographical distance will lead to a tighter time budget for venture capital managers to screen entrepreneurial firms and for post-investment supervision and value-added services. But how does the increased geographical distance affect the investment strategies and investment performance of VCs? This paper presents a large-sample empirical study on these important issues by using the VC data of 21,580 investment rounds from 2000 to 2016 in China.

The study found that the geographical distance between the lead VCF and the IEF is significantly positively correlated with the probability of the IEF's successful IPO or M&A. The reason is that the lead VCFs have adopted a more conservative 'de-risk' strategy when the geographical

distance is farther. For distant investments, VCFs tend to approach with increased caution, adopting more conservative investment strategies. Firstly, VCFs opt for later-stage projects to enhance the likelihood of a successful investment. Secondly, lead VCFs elevate the quality of IEFs by raising their screening standards. Thirdly, they implement a syndication strategy, involving more VCFs in the investment process. Fourthly, a staged financing approach is employed to encourage IEFs to undergo subsequent financing rounds. This conservative investment strategy contributes significantly to improving the probability of a successful exit, whether through IPO or M&A, for the IEFs.

This study extends the previous literature, which has found that increased geographical distance will significantly constrain VC behaviour (Colombo et al., 2019; Cumming & Dai, 2010; Lutz et al., 2013; Tykiová & Schertler, 2014). Based on a large sample of data, we confirm that the investment obstacles caused by the increase in geographical distance can be compensated for by reducing risk preference through improved screening standards and supplemented by syndication and staged financing strategies in order to achieve a higher probability of a successful exit from the IEF. However, the impact of distance on the successful exit rate of VCFs is influenced by some other factors, such as the VCF's experience, economic uncertainty, geopolitical risk and the market development.

The findings of the study could also provide some implications for VCFs, entrepreneurial firms and local government. As the geographical distance increases significantly, the rational choice for VCFs is to invest in the later-stage projects. Therefore, VCFs that focus on investing in an early stage could arrange resident offices in provincial cities with developed high-tech industries besides being in Beijing, Shanghai and Shenzhen. The positive influence of syndication and staged financing implies that, when dealing with distant investments, VCFs can employ these strategies to enhance the success rate. Furthermore, acknowledging the negative moderating influence of factors like economic uncertainty and market conditions on distance, local governments can attract VC inflows by fostering a favourable market environment.

There are also some limitations to the study we have to acknowledge. First, advancements in transportation technology may influence the barrier effect of distance to some extent. This could potentially impact the relationship between distance and VC investment performance. This presents a valuable topic worthy of further investigation. Second, the influence of distance might manifest in the relocation of VCFs and the establishment of branch offices, constituting a subject worthy of investigation. Some factors, including education, style drift and market competition, can exert an influence on VC investments (Cumming et al., 2022; Florian et al., 2022; Li et al., 2022; Lukas & Hans-Peter, 2022). These aspects warrant consideration in the examination of the relationship between distance and investment performance. However, due to the difficulty of data acquisition, this paper hasn't investigated these aspects. Finally, exploring the impact of distance on IPO underpricing or long-term performance represents a valuable area for future research.

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